BIO 005 Human Physiology · Week 11 notes · Dr. Sharilyn Rennie

Blood

Need-to-know blood anatomy, blood functions and plasma proteins, hematocrit, red cells and hemoglobin, erythropoiesis and erythropoietin, red cell breakdown and jaundice, anemias and erythrocytosis, white blood cells, hemostasis and the coagulation cascade, clot limits and anticoagulant drugs, ABO and Rh blood types with hemolytic disease of the newborn, and the hematocrit, hemoglobin and blood typing labs. Week 11 competencies 1 to 12.

To print: these notes as a two-column PDF.

How to use these notes. Read these notes alongside your textbook reading for the week, then use them to review. Each part names where to read in Silverthorn and in OpenStax, and links Dr. Rennie's videos where there is one.

Your textbook: Silverthorn Chapter 17, Blood.
Using OpenStax instead? Anatomy and Physiology 2e, Chapter 18, The Cardiovascular System: Blood.

Need-to-know anatomy

In your textbook: Silverthorn Chapter 17, the parts on plasma and the cellular elements of blood and on blood cell production.

Dr. Rennie's videos for this part: General characteristics of blood, Formed elements and extracellular matrix of blood.

Blood is a fluid connective tissue. Like every connective tissue it has cells and an extracellular matrix. In blood the cells and cell fragments are called the formed elements, and the matrix is a liquid called plasma. An adult has about 4 to 6 liters of blood, which is about 7 to 8% of body weight.

- Blood is thicker (more viscous) than water, because of its cells and proteins.
- Its pH stays between 7.35 and 7.45, slightly alkaline.
- Its temperature is about 38 °C (100.4 °F), slightly above normal body temperature.
- Oxygen-rich blood is bright red. Oxygen-poor blood is dark red, not blue.

What a tube of blood is made of

Part	Share of blood volume	What it is
Plasma (top layer)	About 55%	Straw-colored liquid: about 92% water, 7% proteins, 1% ions, nutrients, gases, hormones and wastes
Buffy coat (thin middle layer)	Less than 1%	White blood cells and platelets
Red blood cells (bottom layer)	About 45%	Erythrocytes, packed with hemoglobin

Formed element	What it is	Normal count	Life span
Red blood cell (erythrocyte)	Biconcave disc with no nucleus or organelles	About 4.7 to 6.1 million per µL in men, 4.2 to 5.4 million in women	About 120 days
White blood cell (leukocyte)	Complete cell with a nucleus; five types	About 4,500 to 11,000 per µL	Hours to years, by type
Platelet (thrombocyte)	Cell fragment broken off a megakaryocyte; no nucleus	About 150,000 to 400,000 per µL	About 7 to 10 days

Reference ranges in these notes are typical values. Each lab prints its own range with the result, and small differences between labs are normal.

Where blood cells are made and removed

Red bone marrow

The tissue where all blood cells are made. (Lymphocytes start here but some finish maturing in the thymus and lymphatic organs.) In adults it fills the flat bones and the ends of long bones: the pelvis, sternum, ribs, vertebrae, skull and the upper ends of the femur and humerus.

Hematopoietic stem cell

The single marrow cell type that gives rise to every blood cell. It splits into two lines: the myeloid stem cell and the lymphoid stem cell.

Spleen

An organ in the upper left abdomen that filters blood. Its macrophages remove old red cells, and it holds about a third of the body's platelets in reserve.

Liver

Makes most plasma proteins, including albumin, fibrinogen and most clotting factors, and clears bilirubin from the blood.

Kidneys

Make erythropoietin, the hormone that drives red cell production.

The vessel wall, for hemostasis

Endothelium

The single layer of flat cells lining every blood vessel. Healthy endothelium is smooth and releases chemicals that keep blood from clotting.

Collagen

Protein fibers in the connective tissue just under the endothelium. Blood normally never touches it. When a vessel tears, exposed collagen is the signal that starts a clot.

Smooth muscle

The muscle layer in the wall of arteries, arterioles and veins. It can contract and squeeze a damaged vessel narrower.

What blood does, plasma proteins and hematocrit

Competencies 1 and 2: Plasma and blood functions; Hematocrit and formed elements.

In your textbook: Silverthorn Chapter 17, the part on plasma and the cellular elements of blood.

Dr. Rennie's videos for this part: Functions of blood, Plasma proteins alpha, beta and gamma globulins, Hematocrit.

Three jobs of blood

Category	What blood does
Transport	Carries O2 from the lungs and CO2 back to them, nutrients from the gut, wastes to the kidneys and liver, and hormones to their targets
Regulation	Buffers pH, carries heat from active muscles and organs to the skin, and holds water in the vessels through the osmotic pull of plasma proteins
Protection	Clots to stop blood loss, and carries white blood cells, antibodies and complement proteins that fight infection

Plasma proteins

Plasma proteins stay in the blood because they are too large to cross most capillary walls. Albumin is about 55 to 60% of plasma protein, globulins about 35 to 38%, and fibrinogen about 4 to 7%. The liver makes almost all of them. Antibodies are the exception: plasma cells, the antibody-making form of B lymphocytes, produce them.

Protein	Made by	Main job
Albumin	Liver	Creates most of the colloid osmotic pressure that holds water inside capillaries; carries fatty acids, bilirubin, some hormones and many drugs
Alpha and beta globulins	Mostly liver	Transport proteins: lipoproteins carry cholesterol and fats, transferrin (a beta globulin) carries iron, others carry hormones and metals
Gamma globulins	Plasma cells	Antibodies (immunoglobulins) that bind foreign antigens
Fibrinogen	Liver	Clotting protein; thrombin turns it into fibrin threads

Plasma also carries clotting factors, complement proteins, enzymes and hormones. When the liver fails, albumin falls, colloid osmotic pressure drops, and fluid leaks out of the capillaries into the tissues as edema.

Serum

The fluid left after blood clots. It is plasma minus fibrinogen and the other clotting factors used up in the clot.

Colloid osmotic pressure

The osmotic pull created by proteins that cannot leave the capillary. It draws water back into the blood.

Hematocrit

Hematocrit (Hct)

The percentage of whole blood volume made up of red blood cells. Normal is about 42 to 52% in men and 37 to 47% in women. Testosterone raises erythropoietin production, which is one reason men run higher.

A hematocrit is a ratio of cells to whole blood, so it changes when either the red cells or the plasma changes. Before calling a value anemia, ask what happened to the plasma.

Result	Possible cause	What changed
Low hematocrit	Anemia	Fewer red cells or smaller red cells
Low hematocrit	Large amounts of IV fluid, late pregnancy	Extra plasma water dilutes the red cells
High hematocrit	Dehydration (vomiting, diarrhea, burns, heavy sweating)	Plasma shrank; the red cell number did not change
High hematocrit	High altitude, chronic lung disease, polycythemia	The body truly made more red cells

Right after a sudden hemorrhage the hematocrit can look normal, because whole blood was lost, cells and plasma together. Over the next hours, fluid moves from the tissues into the plasma and dilutes the remaining red cells, and the hematocrit falls. Over the following days to weeks, erythropoietin raises red cell production and brings the count back up.

Red blood cells and hemoglobin

Competency 3: Erythrocyte structure and hemoglobin.

In your textbook: Silverthorn Chapter 17, the part on red blood cells and hemoglobin.

Dr. Rennie's videos for this part: Erythrocytes (red blood cells), Hemoglobin.

A mature red blood cell does one job: it carries O2 and helps carry CO2. Each is a flexible biconcave disc about 7.5 µm across, thin in the middle and thicker at the rim. As it matures in the marrow it throws out its nucleus and organelles, which leaves it packed with hemoglobin. Every feature of its structure serves gas transport.

Feature	Why it matters
Biconcave shape	Large surface area for its volume, and no part of the cell is far from the surface, so gases diffuse in and out quickly
Flexible membrane held by the protein spectrin	Folds to squeeze single file through capillaries narrower than the cell, then springs back
No nucleus or organelles	More room for hemoglobin, but the cell cannot divide, make new proteins or repair itself, so it wears out in about 120 days
No mitochondria	Makes ATP only by glycolysis, which does not need O2, so the cell does not use up the O2 it carries
Carbonic anhydrase inside	Speeds the reaction of CO2 with water that forms H+ and HCO3−, the main way CO2 travels in blood

Hemoglobin structure

Each hemoglobin (Hb) molecule is built from four protein chains called globins. Adult hemoglobin has two alpha chains and two beta chains. Each chain holds one heme group, a ring with an iron ion (Fe2+) at its center. Each iron binds one O2 molecule, so one hemoglobin molecule can carry four O2.

How hemoglobin carries gases

- More than 98% of the O2 in blood rides on hemoglobin; the rest is dissolved in plasma.
- In the lungs, where PO2 is high, O2 binds the iron and forms oxyhemoglobin (bright red). In the tissues, where PO2 is low, O2 is released and deoxyhemoglobin (dark red) is left.
- Binding is cooperative: once one heme binds O2, the others bind more easily. The reverse is also true, so hemoglobin unloads O2 readily where tissues need it.
- Hemoglobin carries some CO2, bound to the globin chains rather than the iron (carbaminohemoglobin).
- Hemoglobin binds H+, which makes it an important buffer in red cells.
- Carbon monoxide binds heme far more tightly than O2 does, so even small amounts block O2 transport.

Because O2 travels almost entirely on hemoglobin, the amount of hemoglobin in the blood sets how much O2 the blood can carry. A person with half the normal hemoglobin carries about half the O2, even when the lungs work perfectly.

Making red blood cells

Competency 4: Erythropoiesis and its regulation.

In your textbook: Silverthorn Chapter 17, the part on blood cell production, including erythropoietin and iron.

Dr. Rennie's videos for this part: Terminology related to erythrocytes and stages of erythropoiesis, Hormonal regulation of erythropoiesis, Iron.

The body replaces red cells at a rate of millions every second, and the marrow must match production to loss. Too few red cells starve tissues of O2. Too many thicken the blood. A hormone, erythropoietin, keeps the count in range.

Hematopoiesis

The making of all blood cells in red bone marrow.

Erythropoiesis

The making of red blood cells.

Erythropoietin (EPO)

A hormone made mainly by the kidneys that tells the marrow to make more red cells.

Reticulocyte

A young red cell that has just pushed out its nucleus. It still has some ribosomes, which show as a fine network when stained. Reticulocytes are normally about 0.5 to 2.5% of red cells in adults. A rising reticulocyte count shows the marrow is responding.

Hypoxia

Too little O2 reaching the tissues.

Stages of erythropoiesis, step by step

1. A hematopoietic stem cell in red marrow divides and gives rise to a myeloid stem cell.
2. The myeloid stem cell produces a cell committed to the red cell line, the proerythroblast.
3. Proerythroblasts divide and become erythroblasts, which fill with hemoglobin.
4. Each erythroblast pushes out its nucleus and most organelles and becomes a reticulocyte.
5. The reticulocyte leaves the marrow and enters the blood.
6. Within about 1 to 2 days in the blood, it loses its last ribosomes and becomes a mature erythrocyte.

The erythropoietin feedback loop, step by step

1. Something lowers O2 delivery to the tissues: blood loss, anemia, high altitude or lung disease. This is the stimulus.
2. Cells in the kidney tissue around the tubules sense the low O2. A transcription factor called hypoxia-inducible factor switches on the EPO gene.
3. The kidneys release EPO into the blood. The liver adds a small amount.
4. EPO travels to the red bone marrow and binds receptors on red cell precursors.
5. The precursors divide and mature faster, and more reticulocytes enter the blood within a few days.
6. The red cell count and hemoglobin rise, so the blood carries more O2.
7. O2 delivery to the kidney returns to normal, and EPO release falls. The response turns itself off, which is negative feedback.

Note that the kidney senses O2 delivery, not the number of red cells. That is why anything that lowers O2, even with a normal red cell count, raises EPO. People with chronic kidney disease often become anemic because damaged kidneys make too little EPO. Synthetic EPO can correct this, and athletes have misused it to raise their hematocrit, which thickens the blood.

Nutrients the marrow needs

Nutrient	Why it is needed	What a shortage causes
Iron	Sits at the center of every heme; without it, hemoglobin cannot be made	Iron deficiency anemia: small, pale red cells
Vitamin B12	Needed for DNA synthesis in dividing precursors; absorption needs intrinsic factor from the stomach's parietal cells	Large, fragile red cells (megaloblastic anemia); pernicious anemia when intrinsic factor is missing
Folate (folic acid)	Also needed for DNA synthesis in dividing precursors	Large, fragile red cells (megaloblastic anemia)
Amino acids	Build the globin chains	Poor red cell production in protein malnutrition

How the body handles iron

- Iron is absorbed in the small intestine, mostly the duodenum.
- In plasma it travels bound to transferrin, a transport protein.

- It is stored as ferritin (and hemosiderin), mainly in the liver, spleen and bone marrow.
- Free iron is toxic, so the body keeps it bound to a protein almost all the time.
- Most iron for new red cells is recycled from old ones. The body has no regulated way to excrete iron. It loses only small amounts each day, mostly in shed cells, unless it bleeds.
- Because of this, chronic blood loss, such as heavy menstrual bleeding or slow bleeding in the gut, is a common cause of iron deficiency. Pregnancy and growth also raise iron needs.

Red cell breakdown, jaundice and red cell disorders

Competency 5: Red cell destruction and bilirubin.

In your textbook: Silverthorn Chapter 17, the part on red blood cells, including hemoglobin breakdown and red blood cell disorders.

Dr. Rennie's videos for this part: Turnover of RBCs, Anemia, Thalassemia, Sickle cell anemia, Erythrocytosis.

A red cell lasts about 120 days. With no nucleus it cannot repair itself, so its membrane slowly becomes stiff and fragile. Macrophages remove worn cells, mostly in the spleen, with help from the liver and bone marrow. The spleen's narrow passages act as a filter: a flexible young cell squeezes through, and a stiff old one is trapped and eaten.

Red cell breakdown, step by step

1. An aged, stiff red cell is trapped in the spleen and engulfed by a macrophage.
2. Inside the macrophage, hemoglobin is split into globin and heme.
3. Globin is broken down to amino acids, which are reused to build new proteins.
4. Iron is removed from heme. Transferrin carries it to the bone marrow for new hemoglobin, or it is stored as ferritin.
5. The rest of the heme ring is converted to biliverdin, a green pigment, and then to bilirubin, a yellow pigment.
6. Bilirubin enters the plasma and rides to the liver bound to albumin. This form is unconjugated bilirubin, and it is not water soluble.
7. Liver cells attach glucuronic acid to bilirubin (conjugation), which makes it water soluble, and secrete it into bile.
8. Bile carries bilirubin into the small intestine. Bacteria in the large intestine convert it to urobilinogen.
9. Most urobilinogen becomes stercobilin, which gives feces their brown color.
10. A small amount of urobilinogen is reabsorbed into the blood and excreted by the kidneys as urobilin, the yellow pigment of urine.

Jaundice

Jaundice is yellowing of the skin and of the whites of the eyes (sclera) when bilirubin builds up in the blood. Any step in the pathway above can fail, and where it fails tells you which kind of bilirubin builds up.

Type	What goes wrong	Examples
Before the liver (prehepatic)	Red cells break down faster than the liver can clear bilirubin; mostly unconjugated bilirubin rises	Hemolytic anemias, transfusion reactions, hemolytic disease of the newborn
In the liver (hepatic)	Damaged liver cells cannot take up, conjugate or secrete bilirubin; unconjugated and conjugated bilirubin can both rise	Hepatitis, cirrhosis; normal newborn jaundice, because a newborn's liver enzymes are still immature while fetal red cells break down quickly
After the liver (posthepatic)	Bile cannot drain, so conjugated bilirubin backs up into the blood	A gallstone or tumor blocking the bile duct

In posthepatic jaundice, conjugated bilirubin is water soluble, so when it backs up it spills into the urine and turns it dark. Stools turn pale because less pigment reaches the gut. Unconjugated bilirubin is bound to albumin and does not appear in urine.

Terms for describing red cells

Microcytic, macrocytic, normocytic

Smaller than normal, larger than normal, or normal in size.

Mean corpuscular volume (MCV)

The average size of a red cell, reported in femtoliters (fL). Normal is about 80 to 100 fL. Below 80 fL is microcytic, above 100 fL is macrocytic, and 80 to 100 fL is normocytic.

Hypochromic, normochromic

Paler than normal because of less hemoglobin per cell, or normal color.

Hemolysis

Rupture of red cells, which releases hemoglobin into the plasma.

Anisocytosis and poikilocytosis

Red cells of unequal sizes, and red cells of abnormal shapes.

Anemia

Anemia means the blood carries too little O2 because it has too few red cells or too little hemoglobin. The person feels tired and weak, looks pale and gets short of breath with effort. The heart beats faster to move the smaller O2 supply around more often. Anemia has three basic causes: blood loss, too little production, or too much destruction (hemolysis).

Condition	Mechanism	Red cells and MCV
Hemorrhagic anemia	Sudden or slow blood loss	Normal size at first (normal MCV); small and pale (low MCV) if slow loss drains iron
Iron deficiency anemia	Too little iron to make heme, from chronic blood loss, low intake or higher need such as pregnancy	Microcytic and hypochromic (MCV below 80 fL)
B12 or folate deficiency (megaloblastic) anemia	Precursors cannot make DNA fast enough to divide normally	Macrocytic and fragile (MCV above 100 fL)
Anemia of kidney disease	Damaged kidneys make too little EPO	Normal size and color, too few (normal MCV)
Aplastic anemia	The marrow fails, so red cells, white cells and platelets all fall	Usually normal size and color, too few (normal MCV)
Sickle cell anemia	Inherited hemoglobin S (see below)	Rigid crescent shapes (MCV usually normal)
Thalassemia	Inherited loss of globin chain production (see below)	Microcytic and hypochromic (MCV below 80 fL)

Sickle cell anemia

Sickle cell anemia is inherited. A change of one amino acid in the beta globin chain (valine in place of glutamic acid) makes hemoglobin S. When hemoglobin S gives up its O2, the molecules stick together into long rods that bend the red cell into a stiff crescent (sickle) shape.

- Sickled cells jam in small vessels, which blocks blood flow and causes painful crises and tissue damage.
- Sickled cells are fragile and survive only about 10 to 20 days, so the marrow cannot keep up: anemia and jaundice follow.
- Low O2, dehydration and infection make sickling worse.
- It is autosomal recessive. A person with one sickle gene (sickle cell trait) usually has no symptoms, and the trait gives some protection against malaria.

Thalassemia

Thalassemias are inherited disorders in which the genes for alpha or beta globin chains make too little chain or none. With too few normal chains the cell makes too little hemoglobin, so red cells are small, pale and fragile and are destroyed early. Severity depends on how many genes are affected, from a mild carrier state to severe anemia needing regular transfusions.

Erythrocytosis (polycythemia)

Erythrocytosis

An abnormally high red cell mass, which raises the hematocrit. The words erythrocytosis and polycythemia are often used for the same thing; polycythemia vera is one specific cause.

Secondary erythrocytosis

The body makes extra red cells because EPO is high: chronic hypoxia from high altitude, chronic lung disease or heavy smoking, an EPO-secreting tumor, or EPO misuse.

Polycythemia vera

A bone marrow disorder in which the marrow overproduces red cells, often with extra white cells and platelets, without an EPO signal. EPO is low.

Relative erythrocytosis

A high hematocrit from low plasma volume, as in dehydration. The red cell mass is normal.

Extra red cells make blood more viscous. Thicker blood raises resistance to flow, makes the heart work harder, can raise blood pressure and raises the risk of clots.

White blood cells

Competency 6: Leukocytes.

In your textbook: Silverthorn Chapter 17, the parts on the cellular elements of blood and on blood cell production.

Dr. Rennie's videos for this part: Leukocytes and cell lines, Unique characteristics of leukocytes, Terminology associated with leukocytes, cell disorders and neutropenic calculation.

White blood cells (leukocytes) are the only formed elements that are complete cells, with a nucleus and organelles. They do most of their work outside the blood. The bloodstream carries them to infected or injured tissue, where they leave the vessel and go to work.

What makes leukocytes unique

- Diapedesis: they squeeze between the endothelial cells of a capillary wall to leave the blood.
- Amoeboid movement: they crawl through tissue by pushing out parts of the cell.
- Chemotaxis: they follow chemical signals released by damaged tissue, microbes and other immune cells.
- The marrow raises their numbers quickly when the body is under attack.

Cell lines

The hematopoietic stem cell gives rise to two lines. The myeloid stem cell makes red cells, platelets (from giant megakaryocytes), the three granulocytes and monocytes. The lymphoid stem cell makes lymphocytes. Chemical messengers called colony-stimulating factors and interleukins drive white cell production, and thrombopoietin, made mainly by the liver, drives platelet production.

Granulocytes

Neutrophils, eosinophils and basophils. Their cytoplasm has granules that show when stained, and their nuclei have lobes.

Agranulocytes

Lymphocytes and monocytes. Their granules do not show with routine stains.

The five types

From most to least common they are neutrophils, lymphocytes, monocytes, eosinophils and basophils ("Never Let Monkeys Eat Bananas"). A differential count reports each as a percentage of the white cells.

Cell	Share	Main job	Raised in
Neutrophil: granulocyte, nucleus with 3 to 5 lobes	50 to 70%	First responder; engulfs and kills bacteria	Acute bacterial infection, tissue injury, stress
Lymphocyte: agranulocyte, large round nucleus	20 to 40%	B cells become plasma cells that make antibodies; T cells kill infected cells and direct the immune response; NK cells kill virus-infected and cancer cells	Viral infections such as mononucleosis; some leukemias
Monocyte: largest, kidney-shaped nucleus	2 to 8%	Leaves the blood and becomes a macrophage that engulfs microbes and debris	Chronic infections such as tuberculosis, chronic inflammation
Eosinophil: two-lobed nucleus, red-orange granules	1 to 4%	Attacks parasitic worms; helps control allergic reactions	Parasitic worm infections, allergies, asthma
Basophil: dark blue-purple granules	0.5 to 1%	Releases histamine (widens vessels) and heparin (anticoagulant) in inflammation and allergy	Allergic reactions; some leukemias

Terms and disorders

Leukocytosis

A white count above the normal range (above about 11,000 per µL), usually a normal response to infection or inflammation.

Leukopenia

A white count below the normal range (below about 4,500 per µL). It raises the risk of infection.

Neutrophilia, lymphocytosis, eosinophilia

A rise in one type of white cell. The type that rises points to the cause, as in the table above.

Neutropenia

Too few neutrophils, as after chemotherapy or with marrow failure. Bacterial infections can become severe quickly.

Bands and the shift to the left

Bands are young neutrophils with an unsegmented nucleus. More bands than normal (a shift to the left) means the marrow is releasing neutrophils early, usually in acute bacterial infection.

Leukemia

A cancer of a white cell line. The marrow fills with abnormal cells that do not defend well and crowd out normal production, so anemia, bleeding and infections follow even when the white count is very high.

Infectious mononucleosis

A viral infection (Epstein-Barr virus) with many large, atypical lymphocytes in the blood.

Absolute neutrophil count

A percentage alone can mislead, because 60% of a very low white count is still very few neutrophils. The absolute neutrophil count (ANC) turns the differential into a real number.

1. Add the percentage of mature (segmented) neutrophils and the percentage of bands.
2. Multiply the total white count by that sum.
3. Divide by 100. ANC = white count × (% segs + % bands) ÷ 100.
4. Example: white count 2,000 per µL, 40% segs, 5% bands. 2,000 × 45 ÷ 100 = 900 per µL, which is moderate neutropenia.
5. Compare: a normal ANC is about 1,500 to 8,000 per µL. Below 1,500 per µL is neutropenia, 500 to 1,000 per µL is moderate, and below 500 per µL is severe, with a very high risk of infection.

Platelets, hemostasis and the coagulation cascade

Competencies 7 and 8: Hemostasis; Coagulation cascade.

In your textbook: Silverthorn Chapter 17, the parts on platelets and on hemostasis and coagulation.

Dr. Rennie's videos for this part: Platelets (thrombocytes), 3 phases of hemostasis, Phase I of hemostasis, Phase II of hemostasis, Phase III of hemostasis, Intrinsic, extrinsic and common pathway.

Platelets are cell fragments, not whole cells. They break off giant marrow cells called megakaryocytes. A platelet has no nucleus, but it has mitochondria and granules filled with chemical signals such as ADP and serotonin, plus clotting factors and Ca2+. Platelets circulate for about 7 to 10 days.

Hemostasis means stopping bleeding. It happens in three overlapping phases: vascular spasm, platelet plug formation and coagulation. Some books add clot retraction and fibrinolysis as phases IV and V, and the videos use that five-phase count. Phases IV and V are in the next section.

Phase I: vascular spasm

The smooth muscle of a damaged vessel contracts right away, which narrows the vessel and slows blood loss. Direct injury to the muscle, pain reflexes and paracrine signals all trigger it. The signals include endothelin from the damaged endothelium, and serotonin and thromboxane A2 from platelets. The spasm buys time, but it cannot seal the hole by itself.

Phase II: the platelet plug, step by step

1. The injury tears the endothelium and exposes collagen in the vessel wall underneath.
2. Von Willebrand factor (vWF), a plasma protein made by endothelial cells, binds to the exposed collagen.
3. Platelets stick to the vWF and to the collagen through receptors on their membranes. This is platelet adhesion.
4. The stuck platelets activate. They change shape, putting out spiky projections, and release ADP and serotonin from their granules.
5. Activated platelets make thromboxane A2 from fatty acids in their membranes, using the enzyme cyclooxygenase (COX).
6. ADP and thromboxane A2 activate nearby platelets, which turn sticky and join the pile. This is platelet aggregation, a positive feedback loop.
7. Fibrinogen links neighboring activated platelets together, and a loose platelet plug fills the hole.
8. Intact endothelium next to the injury releases prostacyclin and nitric oxide, which block aggregation, so the plug stays at the injury.

Von Willebrand disease, the most common inherited bleeding disorder, shows why step 2 matters. With too little vWF, platelets cannot grip the wound well, and the person bruises easily and bleeds from the nose and gums.

Phase III: coagulation

Coagulation turns the loose platelet plug into a firm clot. The end product is fibrin, a protein thread that weaves

through the plug and traps red cells. Getting there takes a cascade of clotting factors. Most are plasma proteins made by the liver that circulate in an inactive form. Each activated factor is an enzyme that activates the next, so a few molecules at the start produce a large amount of fibrin at the end.

The factors are numbered with Roman numerals in the order they were discovered, not the order in which they act.

Pathway	How it starts	Steps	Speed
Extrinsic (tissue factor pathway)	Damaged tissue outside the blood exposes tissue factor (factor III)	Tissue factor and factor VII, with Ca2+, activate factor X	Fast, seconds
Intrinsic (contact pathway)	Factor XII is activated by exposed collagen, or by glass in a test tube	XII activates XI, XI activates IX, IX with VIII and Ca2+ activates X	Slower, minutes
Common pathway	Both pathways meet at factor X	X with V turns prothrombin into thrombin; thrombin turns fibrinogen into fibrin	Final steps of both

The common pathway, step by step

1. Activated factor X joins factor V and Ca2+ on the surface of activated platelets. Together they form prothrombin activator.
2. Prothrombin activator converts prothrombin (factor II) into thrombin.
3. Thrombin cuts fibrinogen (factor I), a soluble plasma protein, into fibrin pieces that join end to end into long threads.
4. Thrombin activates factor XIII, which cross-links the fibrin threads into a strong mesh.
5. The mesh traps red cells and platelets, and the plug becomes a stable clot.
6. Thrombin also activates more platelets and factors V, VIII and XI. This positive feedback speeds the whole cascade, so clotting builds quickly once it starts.

Separate intrinsic and extrinsic pathways are a simplified model that comes from lab tests. In the body, clotting starts mainly through tissue factor, and the first thrombin then switches on the intrinsic factors, which keep the clot growing. Both are needed: people missing factor VIII or IX (hemophilia) bleed even though their extrinsic pathway works.

Thrombin

The enzyme at the center of clotting. It makes fibrin, activates platelets and speeds up the cascade.

Fibrin

The insoluble protein mesh of a clot, made from fibrinogen.

Calcium (Ca2+, factor IV)

Needed at several steps of all three pathways. Blood collection tubes with citrate or EDTA bind Ca2+, which is why the blood in them does not clot.

Vitamin K

A fat-soluble vitamin the liver needs to make working factors II, VII, IX and X, and the anticlotting proteins C and S. It comes from leafy green vegetables and from bacteria in the large intestine. Newborns have very little, so they routinely receive vitamin K at birth.

Limiting and dissolving a clot

Competency 9: Clot limitation and fibrinolysis.

In your textbook: Silverthorn Chapter 17, the part on hemostasis and coagulation, including anticoagulants and fibrinolysis.

Dr. Rennie's videos for this part: Phases IV and V of hemostasis, Lab test related to coagulation, Anticoagulants, Terminology, Disorders related to clotting.

A clot must stay at the wound. A clot that spreads, or that forms inside an intact vessel, can block blood flow to the heart, brain or lungs. The body limits clotting in several ways and removes the clot once the vessel has healed.

Built-in brakes on clotting

Brake	Comes from	What it does
Antithrombin (antithrombin III)	Plasma protein made by the liver	Inactivates thrombin and factor X, along with other factors; heparin makes it act much faster
Protein C, helped by protein S	Plasma proteins made by the liver; need vitamin K	Thrombin bound to thrombomodulin on healthy endothelium activates protein C, which shuts down factors V and VIII
Prostacyclin and nitric oxide	Intact endothelium	Block platelet aggregation next to the injury
Blood flow	Moving blood	Washes activated factors away and dilutes them

Phases IV and V: clot retraction and fibrinolysis, step by step

1. Within about 30 to 60 minutes, platelets in the clot contract using actin and myosin.
2. The contraction pulls the fibrin threads and the torn vessel edges closer together and squeezes serum out of the clot. This is clot retraction.
3. Platelet-derived growth factor, released by platelets, stimulates smooth muscle cells and fibroblasts to repair the vessel wall, and new endothelium grows over it.
4. While the clot formed, an inactive plasma protein called plasminogen was trapped inside it.
5. Endothelial cells release tissue plasminogen activator (tPA), which converts plasminogen into plasmin.
6. Plasmin, an enzyme, digests fibrin, and the clot dissolves. This is fibrinolysis.
7. Pieces of cross-linked fibrin, called D-dimers, enter the blood.

Drugs that act on clotting

Each drug targets a step you have already seen, so you can predict what it does. All of them raise the risk of bleeding, which is the main side effect to watch for.

Drug	What it acts on	Result
Heparin	Binds antithrombin and makes it far more active, so thrombin and factor X are inactivated	Works within minutes; stops clots from growing but does not dissolve existing clots
Warfarin	Blocks the recycling of vitamin K in the liver	The liver makes nonworking factors II, VII, IX and X; takes several days to work, as the old factors clear
Aspirin	Permanently blocks cyclooxygenase in platelets, so they cannot make thromboxane A2	Less platelet aggregation, lasting for the life of the platelet (about 7 to 10 days), because a platelet has no nucleus to make new enzyme
Clopidogrel	Blocks the ADP receptor on platelets	Less platelet aggregation
tPA (alteplase)	Converts plasminogen into plasmin	Dissolves an existing clot, as in some strokes and heart attacks

Warfarin and vitamin K work against each other. A patient on warfarin who suddenly eats far more leafy greens takes in extra vitamin K, which pushes against the drug, and clotting speeds back up. Vitamin K reverses warfarin's effect for the same reason. Heparin has its own reversal agent, protamine.

Lab tests related to coagulation

Test	What it measures	Used for
Prothrombin time (PT) and INR	How fast plasma clots through the extrinsic and common pathways; the INR is the PT standardized so results compare between labs	Following warfarin, since factor VII has the shortest half-life of the vitamin K factors and falls first; liver disease; vitamin K deficiency
Activated partial thromboplastin time (aPTT)	How fast plasma clots through the intrinsic and common pathways	Following heparin; hemophilia
Platelet count	Number of platelets per µL	Thrombocytopenia, bleeding risk
D-dimer	Fibrin breakdown pieces in the blood	A high level means clots are forming and breaking down somewhere, as with a deep vein clot or pulmonary embolism

A longer PT or aPTT than normal means the blood takes longer to clot. A normal INR is about 1, and a common target on warfarin is 2 to 3. That is the goal when a patient is on an anticoagulant, and a warning sign when they are not.

Clotting terms and disorders

Thrombus

A clot that forms inside an unbroken vessel.

Embolus

A clot, or a bubble of air or fat, that travels in the blood until it lodges in a smaller vessel. The blockage is an embolism.

Deep vein thrombosis (DVT)

A thrombus in a deep vein, usually in the leg, favored by slow blood flow such as long bed rest or long flights. If it breaks loose it can lodge in the lungs as a pulmonary embolism.

Thrombocytopenia

A platelet count below about 150,000 per µL. Very low counts cause pinpoint bleeds in the skin (petechiae) and easy bleeding.

Thrombocytosis

A platelet count above the normal range, which can raise clot risk.

Hemophilia

Inherited lack of a clotting factor: factor VIII in hemophilia A, factor IX in hemophilia B. Both genes are on the X chromosome, so almost all patients are male. Bleeding into joints and muscles is typical.

Vitamin K deficiency and liver disease

Both leave the liver unable to make enough working clotting factors, so the PT rises and the patient bleeds.

Disseminated intravascular coagulation (DIC)

Widespread clotting in small vessels, often triggered by severe infection, trauma or childbirth complications. It uses up platelets and clotting factors, so the patient then bleeds from many sites.

Blood types, transfusion and the blood labs

Competencies 10, 11 and 12: ABO and Rh blood types; Hematocrit and hemoglobin determination; Blood typing.

In your textbook: Silverthorn Chapter 17, the part on red blood cells; antibodies are covered in the immune system chapter.

Dr. Rennie's videos for this part: Human blood groups, Antigens, agglutinogens, agglutinins, Most dangerous RBC antigens, Preformed antibodies, Universal donor, Universal recipient, Hemolytic transfusion reaction, Hemolytic disease of the newborn and RhoGAM, Interpretation of blood typing.

Your blood type is set by antigens on the surface of your red cells. More than 40 blood group systems are known, but the ABO and Rh antigens cause the most dangerous reactions, so they are matched for every transfusion.

Antigen (agglutinogen)

A marker on the red cell surface, a glycoprotein or glycolipid, that an antibody can bind.

Antibody (agglutinin)

A plasma protein (gamma globulin) that binds one specific antigen.

Agglutination

Clumping of red cells. Each antibody has two or more binding sites (anti-A and anti-B are mostly IgM, which has ten), so one antibody can link several red cells together.

Preformed antibodies

Anti-A and anti-B appear in the first months of life without any exposure to foreign blood, probably in response to look-alike antigens on gut bacteria and food. A person makes antibodies against whichever of A and B they lack.

The ABO groups

Type	Antigens on red cells	Antibodies in plasma	Can receive red cells from
A	A	Anti-B	A, O
B	B	Anti-A	B, O
AB	A and B	Neither	A, B, AB, O (universal recipient)
O	Neither	Anti-A and anti-B	O only (universal donor)

To predict a reaction, ask one question: does the recipient's plasma have an antibody against an antigen on the donor's red cells? If yes, the donor cells agglutinate. Type O red cells carry no A or B antigen, so no ABO antibody attacks them. Type AB plasma has no anti-A or anti-B, so it attacks no donor's red cells.

For plasma transfusions the rule flips, because now the donor's antibodies are what matter. Type AB plasma, with no ABO antibodies, can be given to anyone, and type O plasma only to type O patients.

The Rh factor

Rh status refers mainly to the D antigen. Rh positive people have it, and most people are Rh positive. Rh negative people lack it, but unlike ABO antibodies, anti-D is not preformed. An Rh negative person makes anti-D only after exposure to Rh positive red cells, through a transfusion or a pregnancy. A first mismatched transfusion usually causes no immediate reaction but sensitizes the person, and a second one can cause a severe reaction.

Counting both systems, O negative red cells are the universal donor and AB positive patients are the universal recipient. Other blood group antigens still exist, so donated blood is crossmatched against the recipient's blood whenever time allows.

Hemolytic transfusion reaction, step by step

1. Mismatched donor red cells enter the recipient, for example type A cells into a type O patient.
2. The recipient's preformed anti-A antibodies bind the A antigens on the donor cells.
3. The antibodies link donor cells into clumps (agglutination), which can block small vessels.
4. The bound antibodies activate complement, which punches holes in the donor cells, and the cells burst (hemolysis) inside the vessels.
5. Free hemoglobin floods the plasma. It can injure the kidneys and spills into the urine, turning it dark.
6. Heme from the burst cells is broken down to bilirubin, so jaundice can follow.
7. The patient may develop fever, chills, back pain and a drop in blood pressure. This is why transfused patients are watched closely, especially in the first minutes.

Hemolytic disease of the newborn, step by step

1. An Rh negative mother carries an Rh positive fetus, who inherited the D antigen from the father.
2. During the first pregnancy, the two blood supplies stay mostly separate, so this baby is usually not affected.
3. At delivery, or with a miscarriage, abortion or abdominal trauma, some fetal red cells enter the mother's blood.
4. The mother's immune system makes anti-D antibodies and memory cells. She is now sensitized.
5. In a later pregnancy with another Rh positive fetus, her anti-D antibodies, which are IgG and small enough to cross the placenta, enter the fetal blood.
6. The antibodies bind the fetal red cells, which are destroyed.
7. The fetus becomes anemic, and its marrow releases immature erythroblasts into the blood (erythroblastosis fetalis).
8. After birth, without the placenta to clear it, bilirubin climbs, causing severe jaundice and a risk of brain damage (kernicterus).

RhoGAM is an injection of anti-D antibodies given to an Rh negative mother, usually around 28 weeks of pregnancy and again within 72 hours after she delivers an Rh positive baby. She also gets a dose after a miscarriage, abortion, abdominal trauma or amniocentesis. The injected antibodies coat any Rh positive fetal cells in her blood and clear them before her own immune system can respond, so she never becomes sensitized. It works only before sensitization. It cannot remove antibodies she has already made.

An ABO mismatch between mother and baby can also cause hemolytic disease, but it is usually mild, because most anti-A and anti-B antibodies are IgM, too large to cross the placenta.

Lab: hematocrit, step by step

1. Wear gloves and eye protection, and treat every blood sample as possibly infectious.
2. Fill a heparinized capillary tube about two-thirds to three-quarters full with blood from a finger stick or a well-mixed sample.
3. Seal the dry end of the tube with clay.
4. Place the tube in a microhematocrit centrifuge with the sealed end facing out, and spin it for a few minutes.
5. Red cells pack at the bottom, the buffy coat forms a thin band above them, and plasma rises to the top.
6. Measure the height of the red cell column and the height of the whole column (red cells, buffy coat and plasma), or read the percentage directly on a hematocrit reader.
7. Calculate: hematocrit = red cell column height ÷ total column height × 100.
8. Example: red cells 21 mm, total 50 mm. 21 ÷ 50 × 100 = 42%, normal for a woman and at the low end of normal for a man.

Lab: hemoglobin

Hemoglobin is reported in grams per deciliter (g/dL). Normal is about 13.5 to 18 g/dL in men and 12 to 16 g/dL in women. In the lab it is measured with a hemoglobinometer, which reads how much light a blood sample absorbs, or with a color comparison scale.

A quick check on any result: the hematocrit is usually about three times the hemoglobin, so a hemoglobin of 14 g/dL predicts a hematocrit near 42%. If the two do not fit, recheck the sample. Compare each result with the reference range for the person's sex. A low hematocrit with a low hemoglobin suggests anemia. A high hematocrit in a dehydrated person suggests lost plasma rather than extra cells.

Lab: blood typing, step by step

1. Label three spots on a slide or typing tray: anti-A, anti-B and anti-D (anti-Rh).
2. Place a drop of blood, or simulated blood, on each spot.
3. Add a drop of anti-A serum to the first spot, anti-B to the second and anti-D to the third.
4. Mix each spot with a clean stick, using a new stick for each so the sera do not mix.
5. Rock the slide gently and look at each spot. Clumped, grainy blood is a positive reaction. Smooth, even blood is negative.
6. Record which sera caused clumping. Clumping means the red cells carry that antigen.

In this test the antibodies come from the typing serum, not from the person being typed. So clumping with anti-A means the cells have the A antigen. Do not mix this up with a transfusion reaction, where the antibodies come from the recipient's plasma.

Anti-A	Anti-B	Anti-D	Blood type	Can receive red cells from
Clumps	Smooth	Clumps	A positive	A+, A-, O+, O-
Clumps	Smooth	Smooth	A negative	A-, O-
Smooth	Clumps	Clumps	B positive	B+, B-, O+, O-
Smooth	Clumps	Smooth	B negative	B-, O-
Clumps	Clumps	Clumps	AB positive	All eight types
Clumps	Clumps	Smooth	AB negative	AB-, A-, B-, O-
Smooth	Smooth	Clumps	O positive	O+, O-
Smooth	Smooth	Smooth	O negative	O- only

An Rh negative person receives only Rh negative red cells whenever possible, so that no anti-D is made. Rh positive people can receive either, because they make no anti-D.